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Application of Relative Weight Method for Groundwater Quality Assessment in and around Cement Industrial Corridor at Kadapa District, Andhra Pradesh

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Abstract

Groundwater is a vital resource for most of the developmental activities. It is to be noted that the demand for groundwater is increasing due to paucity of surface water and recurrent failures of monsoons. The increasing demand for groundwater causes water level to decline and water quality to deteriorate. The present study intended to determine ground water quality index (WQI) using arithmetic weight methods to establish groundwater quality assessment in and around Yerraguntla, Y.S.R District at Andhra Pradesh. The calculation of WQI for various parameters such as pH, electrical conductivity (EC), total dissolved solids (TDS), total hardness (TH), and major cations and anions. Most of the groundwater samples are exceeding the permissible limits with respect to pH, EC, TDS, TH, nitrates and fluoride. This classification reveals that most of the groundwater is not suitable for drinking purpose. This is clear from this study that groundwater quality is very poor and proper monitoring and management techniques are needed to maintain and improve water quality standards and health of local community.